Candidate Seat No__________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (IT) Examination

January 2023

IT351 Design & Analysis of Algorithms

Date: 04/01/2023, Wednesday   Time: 01:30 p.m. To 4:30 p.m.   Maximum Marks: 70

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1 Do as directed.

1) What is an algorithm? Which are the two ways to analyze the algorithm? **[01]**

2) Find out total number of primitive operations for following code. **[02]**

```
int Array_Min(A, N)
{     current_Min ← A[0]
       for i ← 1 to n − 1 do
         {
            if current_Min > A[i] then
                  current_Min ← A[i]
            i ← i+1
         }
return current_Min
}
```

3) Consider that you have infinite number of coins for the given coin set {1, 5, 8, 25, 50, 100}. You have to give N rupees to your friend with minimum number of coins. Find out two different amount in the range of (1,100) for which the greedy algorithm does not provide optimal solution. Provide set of denominations for each amount. **[02]**

4) Why Greedy algorithms do not provide globally optimized solutions for all the problems? Justify with example. **[02]**

Q-2 (a) Write recurrence relation for the Merge sort algorithm. Discuss best case, average case and worst case performance of the Merge sort algorithm. **[04]**

Q-2 (b) Attempt following Questions (Any Two). **[10]**

1) Discuss general structure of Greedy Algorithm.

2) Assigning 4 people to 4 jobs so that the total cost is minimized. Each person does one job and each job is assigned to one person. Use branch and bound method to solve the assignment problem (Minimization) with the following cost matrices.

	Job 1	Job 2	Job 3	Job 4
Person a	9	2	7	8
Person b	6	4	3	7
Person c	5	8	1	8
Person d	7	6	9	4

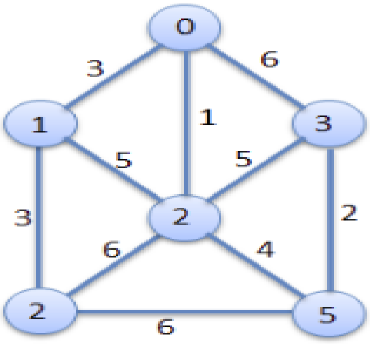
3) Consider there are n jobs, S={1, 2, …, n}, each job i has a deadline $D_i \geq 0$ and a profit $P_i \geq 0$. It is also given that every job takes a single unit of time, so the minimum possible deadline for any job is 1. Using greedy strategy, maximize the total profit if only one job can be scheduled at a time.

i	1	2	3	4	5
$\mathbf{P_i}$	20	15	10	5	1
$\mathbf{D_i}$	2	2	1	3	3

Q-3 (a) Compare and contrast Backtracking and Branch & Bound design technique. **[04]**

Q-3 (b) **Attempt following Questions (Any Two).** **[10]**

1) Generate Minimum Spanning Tree (MST) for below Graph using Prim's algorithm. Consider '0' as a source vertex.



2) Define N-queen problem. Construct pruned state space tree for 5 queen problem for the first solution.

3) What is Divide and Conquer approach? Derive best case, worst case and average case analysis of Binary Search algorithm in terms of Big Oh notation.

SECTION – II

Q - 4 Do as directed.

1) Arrange the following terms in increasing order of their time complexity for large values of n. **[01]**

$O(n^{(3/2)})$, $O(n)$, $O(1)$, $O(\sqrt{n})$, $O(n!)$, $O(\log(\log n))$, $O(2^n)$, $O(n\log n)$

2) Let X be a problem that belongs to the class NP. Then which one of the following is TRUE? **[01]**

(A) There is no polynomial time algorithm for X.
(B) If X can be solved deterministically in polynomial time, then P = NP
(C) If X is NP-hard, then it is NP-complete.
(D) X may be undecidable.

3) List out applications of String Matching algorithms. **[01]**

4) What happens in time complexity and space complexity when a top-down approach of dynamic programming is applied to any problem? **[01]**

5) Define : P problem, NP problem, NP-Complete problem **[03]**

Q-5 (a) Is Dynamic Programming a Top-Down or Bottom-Up technique? Why? Explain with an example. **[04]**

Q-5 (b) Attempt following Questions (Any Two). [10]

1) Working modulo q = 11, how many spurious hits does the Rabin-Karp matcher encounter in the text T = 3141592653589793 when looking for the pattern P = 26?

2) What is Asymptotic Notation? Explain O, Ω, θ notation with a suitable example.

3) Let S = {a, b, c, d, e, f, g} be a collection of objects with benefit-weight values as follows: a: (12,4), b: (10,6), c: (8,5), d: (11,7), e: (14,3), f: (7,1), g: (9,6). What is an optimal solution to 0/1 knapsack problem using dynamic programming assuming the knapsack can hold objects with total weight 18?

Q-6 (a) State Master Theorem and Solve following recurrence using Master Theorem. **[04]**

$$\mathbf{T(n) = 7T(n/3) + n^2}$$

Q-6 (b) Attempt following Questions (Any Two). [10]

1) Find Upper bound of following recurrence equation using Recurrence Tree Method.

$$\mathbf{T(n) = 2T(n/2) + c\,n^2}$$

2) What are the benefits of using Dynamic programming approach over greedy approach? Explain in Detail.

3) Define Longest common subsequence problem. Find the length of the subsequence for X (0 1 1 1 0 1 1) and Y (1 1 1 1 0 0 1).

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